TradingAgents: Multi-Agents LLM Financial Trading Framework

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Significant progress has been made in automated problem-solving using societies of agents powered by large language models (LLMs). In finance, efforts have largely focused on single-agent systems handling specific tasks or multi-agent frameworks independently gathering data. However, the multi-agent systems' potential to replicate real-world trading firms' collaborative dynamics remains underexplored. `TradingAgents` proposes a novel stock trading framework inspired by trading firms, featuring LLM-powered agents in specialized roles such as fundamental analysts, sentiment analysts, technical analysts, and traders with varied risk profiles. The framework includes Bull and Bear researcher agents assessing market conditions, a risk management team monitoring exposure, and traders synthesizing insights from debates and historical data to make informed decisions. By simulating a dynamic, collaborative trading environment, this framework aims to improve trading performance. Detailed architecture and extensive experiments reveal its superiority over baseline models, with notable improvements in cumulative returns, Sharpe ratio, and maximum drawdown, highlighting the potential of multi-agent LLM frameworks in financial trading. TRADINGAGENTS is available at https://github.com/TauricResearch/TradingAgents.

1. Introduction

Autonomous agents leveraging Large Language Models (LLMs) present a transformative approach to decision-making by replicating human processes and workflows across various applications. These systems enhance the problem-solving capabilities of language agents by equipping them with tools and enabling collaboration with other agents, effectively breaking down complex problems into manageable components (Havrilla et al., 2024; Park et al., 2023; Talebirad and Nadiri, 2023; Tang et al., 2024). One prominent application of these autonomous frameworks is in the financial market—a highly complex system influenced by numerous factors, including company fundamentals, market sentiment, technical indicators, and macroeconomic events.

Traditional algorithmic trading systems often rely on quantitative models that struggle to fully capture the complex interplay of diverse factors. In contrast, LLMs excel at processing and understanding natural language data, making them particularly effective for tasks that require textual comprehension, such as analyzing news articles, financial reports, and social media sentiment. Additionally, deep learning-based trading systems often suffer from low explainability, as they rely on hidden features that drive decision-making but are difficult to interpret. Recent advancements in multi-agent LLM frameworks for finance have shown significant promise in addressing these challenges. These frameworks create explainable AI systems, where decisions are supported by evidence and transparent

*Tauric Research Organization: https://tauric.ai reasoning (Li et al., 2023a; Wang et al., 2024c; Yu et al., 2024), demonstrating the potential in financial applications.

Despite their potential, most current applications of language agents in the financial and trading sectors face two significant limitations:

Lack of Realistic Organizational Modeling: Many frameworks fail to capture the complex interactions between agents that mimic the structure of real-world trading firms (Li et al., 2023a; Wang et al., 2024c; Yu et al., 2024). Instead, they focus narrowly on specific task performance, often disconnected from the organizational workflows and established human operating procedures proven effective in trading. This limits their ability to fully replicate and benefit from real-world trading practices.

Inefficient Communication Interfaces: Most existing systems use natural language as the primary communication medium, typically relying on message histories or an unstructured pool of information for decision-making (Park et al., 2023; Qian et al., 2024). This approach often results in a "telephone effect", where details are lost, and states become corrupted as conversations lengthen. Agents struggle to maintain context and track extended histories while filtering out irrelevant information from previous decision steps, diminishing their effectiveness in handling complex, dynamic tasks. Additionally, the unstructured pool-of-information approach lacks clear instructions, forcing logical communication and information exchange between agents to depend solely on retrieval, which disrupts the relational integrity of the data.

In this work, we address these key limitations of existing models by introducing a system that overcomes these challenges. First, our framework bridges the gap by simulating the multi-agent decision-making processes typical of professional trading teams. It incorporates specialized agents tailored to distinct aspects of trading, inspired by the organizational structure of real-world trading firms. These agents include fundamental analysts, sentiment/news analysts, technical analysts, and traders with diverse risk profiles. Bullish and bearish debaters evaluate market conditions to provide balanced recommendations, while a risk management team ensures that exposures remain within acceptable limits. Second, to enhance communication, our framework combines structured outputs for control, clarity, and reasoning with natural language dialogue to facilitate effective debate and collaboration among agents. This hybrid approach ensures both precision and flexibility in decision-making.

We validate our framework through experiments on historical financial data, comparing its performance against multiple baselines. Comprehensive evaluation metrics, including cumulative return, Sharpe ratio, and maximum drawdown, are employed to assess its overall effectiveness.

2. Related Work

2.1. LLMs as Financial Assistants

Large Language Models (LLMs) are applied in finance by fine-tuning on financial data or training on financial corpora. This improves the model's understanding of financial terminology and data, enabling a specialized assistant for analytical support, insights, and information retrieval, rather than trade execution.

Fine-Tuned LLMs for Finance

Fine-tuning enhances domain-specific performance. Examples include PIXIU (FinMA) (Xie et al., 2023), which fine-tuned LLaMA on 136K finance-related instructions; FinGPT (Touvron et al., 2023; Yang et al., 2023b), which used LoRA to fine-tune models like LLaMA and ChatGLM with about 50K finance-specific samples; and Instruct-FinGPT (Zhang et al., 2023a), fine-tuned on 10K instruction samples from financial sentiment analysis datasets. These models outperform their base versions and other open-source LLMs like BLOOM and OPT (Zhang et al., 2022) in finance classification tasks, even surpassing BloombergGPT (Wu et al., 2023) in several evaluations. However, in generative tasks, they perform similarly or slightly worse than powerful general-purpose models like GPT-4, indicating a need for more high-quality, domain-specific datasets.

Finance LLMs Trained from Scratch

Training LLMs from scratch on finance-specific corpora aims for better domain adaptation. Models like BloombergGPT (Wu et al., 2023), XuanYuan 2.0 (Zhang et al., 2023b), and Fin-T5 (Lu et al., 2023) combine public datasets with finance-specific data during pretraining. BloombergGPT, for instance, was trained on both general and financial text, with proprietary Bloomberg data enhancing its performance on finance benchmarks. These models outperform general-purpose counterparts like BLOOM-176B and T5 in tasks such as market sentiment classification and summarization. While they may not match larger closed-source models like GPT-3 or PaLM (Chowdhery et al., 2022), they offer competitive performance among similar-sized open-source models without compromising general language understanding.

In summary, finance-specific LLMs developed through fine-tuning or training from scratch show significant improvements in domain-specific tasks, underscoring the importance of domain adaptation and the potential for further enhancements with high-quality finance-specific datasets.

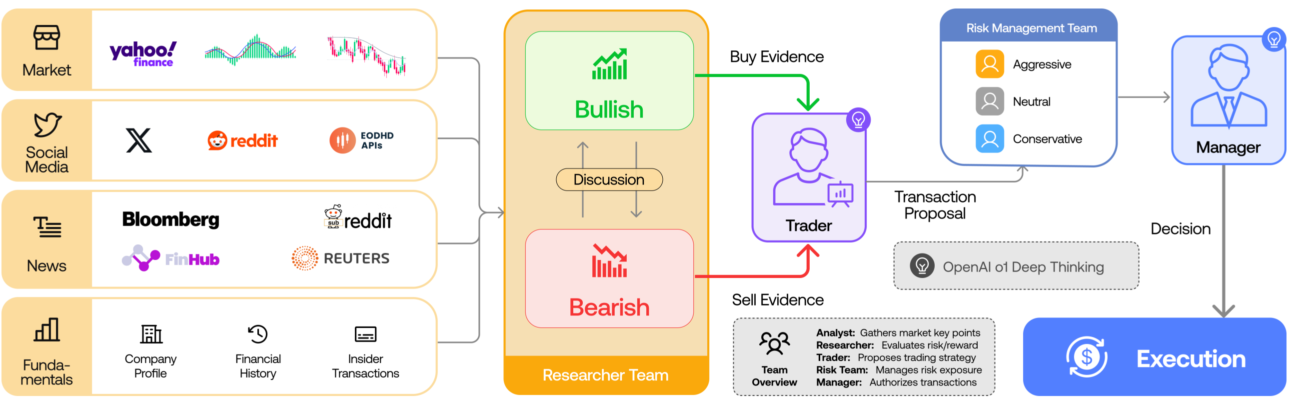

Fig. 1: TradingAgents Overall Framework Organization. **I. Analysts Team**: Four analysts concurrently gather relevant market information. **II. Research Team**: The team discusses and evaluates the collected data. **III. Trader**: Based on the researchers' analysis, the trader makes the trading decision. **IV. Risk Management Team**: Risk guardians assess the decision against current market conditions to mitigate risks. **V. Fund Manager**: The fund manager approves and executes the trade.

2.2. LLMs as Traders

LLMs act as trader agents making direct trading decisions by analyzing external data like news, financial reports, and stock prices. Proposed architectures include news-driven, reasoning-driven, and reinforcement learning (RL)-driven agents.

News-Driven Agents

News-driven architectures integrate stock news and macroeconomic updates into LLM prompts to predict stock price movements. Studies evaluating both closed-source models (e.g., GPT-3.5, GPT-4) and open-source LLMs (e.g., Qwen (Bai et al., 2023), Baichuan (Yang et al., 2023a)) in financial